Hadi Lashkari Ghouchani

Software Engineer | Developer Platform & AI Agents

@
github.com/hadilq
hadilq.com
x.com/@hadilq
Toronto, Canada
linkedin.com/in/hadilq

Summary

Software engineer with 21+ years of development experience and 11 years in industry, the last several focused on developer platforms: CI/CD, build systems, reproducible environments, and AI coding agents that other engineers depend on every day. Deep Kotlin expertise, working Python, and a background in algorithms and applied machine learning.
- Treats developers as customers: gathers and analyzes their feedback daily and proactively fixes what slows them down
- Built a multi-agent CI workflow that turns tickets into merge requests, with coder and reviewer agents in a loop
- Owns CI/CD as production infrastructure: pipeline reliability, runners on AWS EC2, caching, security scanning
- Shortens the path from ticket to merged change: tiered merge request checks, merge trains, faster tests and builds
- Makes environments reproducible and isolated with Nix flakes and task-specific containers
- Served as the build and tooling resource for 50+ engineers; ran RFC and knowledge-sharing practices
- Designs for verifiability: open-source work holds three languages to one behaviour via shared conformance tests
- Taught algorithms and programming at university level
Looking to own how code gets written, validated, and shipped by people and agents.

Experience

Senior Mobile Developer

at Bell Canada | April, 2026 - Present | Toronto, Canada
Bell Canada is a Canadian telecommunications company

Mission:

- Treat the developers who use CI/CD as my customers: collect and analyze their feedback, ship improvements to their experience daily, and proactively find and fix problems before they are reported
- Built a multi-agent CI workflow that turns Jira tickets into merge requests: a planner agent scopes the ticket, then coder and reviewer agents iterate in a loop until the change is ready to commit for human review
- Took ownership of GitLab CI/CD for Android and iOS and reversed a declining pipeline job success rate, raising it from 18% to 31% (a 72% relative improvement) within six months
- Restructured merge request pipelines into fast checks, required heavy jobs, and optional jobs, and strengthened automated code verification, shortening feedback loops and raising confidence before merge
- Designed and implemented trunk-based development pipelines across multiple repositories, automating release-branch creation, cherry-picks, publishing, and release-candidate promotion
- Run the AWS EC2 instances behind CI/CD as production infrastructure, tuning them for reliability and performance
- Made builds and emulator runs reproducible by wrapping them in Nix flakes, and migrated GitLab jobs to custom, task-specific Linux containers for performance and isolation
- Automated DTO and mapper generation from APIDog API specs in CI, eliminating hand-written boilerplate from developer merge requests
- Reduced GitLab SAST security findings to zero across multiple projects
- Pinned dependencies, tuned CI/CD caching, and established architecture and release-readiness practices

Technologies:

- Android, Kotlin Multiplatform, Gradle Plugin
- iOS, Swift, Xcode, xcodebuild
- CI/CD, GitLab CI, GitLab SAST, LLM coding agents, Podman, Nix, AWS EC2

Senior Mobile Developer

at Block Inc. - Bitkey project(via Touchlab) | November, 2025 - Feb, 2026 | Toronto, Canada
Block, Inc. is an American technology company and a financial services provider for consumers and merchants. The Bitkey is a project in Block Inc. that I directly work on. Bitkey is a digital wallet for bitcoin, available in the U.S. and 95 other countries.

Mission:

- Worked on multi backup project
- Fixed flaky tests, including unit and functional tests by solving multiple race conditions and state management problems
- Worked on the new onboarding flow

Technologies:

- Kotlin Multiplatform/KMP, Compose Multiplatform/CMP, Gradle Plugin
- iOS, Swift, Xcode
- Rust binding, UniFFI
- Ktor, Anvil, SqlDelight, Kotest
- Github, Linear, Notion, BuildKite, etc.

Lead Android Developer

at CIBC (via Vision Talent) | March, 2025 - November 2025 | Toronto, Canada
The Canadian Imperial Bank of Commerce is a Canadian multinational banking and financial services corporation headquartered at CIBC Square in the Financial District of Toronto, Ontario.

Mission:

- Rewriting features in the newly developing CIBC mobile apps
- Improved the shared CI pipeline build-time by almost 50%, mostly by using build caches
- Initiated and Implemented memory management by applying structural DI scopes to release resources, ViewModels(in MVVM architecture), and their state
- Working on remote FeatureFlag functionality
- Migrated a Confluence's scraping script to use Selenium

Technologies:

- Kotlin Multiplatform/KMP, Android, Compose Multiplatform/CMP, Gradle Plugin, code generator
- iOS, Swift, Xcode, SwiftUI
- Ktor, Koin, Compose Navigation, etc.
- Atlassian Jira, Atlassian Confluence, Github, Jenkins, etc.

Software Engineer -> Productivity Engineer

at Bell Canada (via MobileLIVE) | Aug, 2022 - Nov, 2024 | Toronto, Canada
Promoted to Productivity Engineer.

Mission:

- Resolved VPN network proxy and self-signed certificate issues by developing an automated installation script and Compose Multiplatform app, resulting in a 164% increase in build count on the Gradle Enterprise(Develocity) dashboard.
- Served as the primary build and tooling resource for 50+ Android developers across teams, providing training, support, and mentoring.
- Initiated and implemented Gradle Convention Plugins shared across teams to reduce code duplication and errors. Established Single Source of Truth (SSOT) for external library versions, significantly reducing runtime crashes caused by version conflicts.
- Spearheaded the adoption of Merge Trains and Merge Request Pipelines in GitLab, creating a Proof of Concept (POC) workflow that demonstrably improved code quality and streamlined the integration process.
- Optimized test execution, reducing incremental runtime by 52% and improving overall development efficiency.
- Implemented a custom No-Framework dependency injection solution across multiple app and library repositories, enhancing modularity, testability, and maintainability while minimizing build times by avoiding code generation.
- Integrated Detekt to verify code quality in the merge request's pipelines.

Additional Technologies:

- Develocity(Gradle Enterprise), Gradle Convention Plugins, Detekt
- Kotlin Multiplatform, Compose Multiplatform

Software Engineer

at Bell Canada (via MobileLIVE) | Apr. 2021 - Aug. 2022 | Toronto, Canada
MobileLIVE: A Canadian tech-service provider specializing in digital transformation solutions. Bell Canada: Canada's largest telecommunications company, offering mobile services (Bell Mobility, Virgin Mobile), television (Bell Satellite TV, Bell Fibe TV), and internet services across the country.

Mission:

- Implemented a Request for Comments (RFC) process for technical decision-making, streamlining the evaluation and adoption of new ideas across the organization.
- Led regular knowledge-sharing discussions across the team on SOLID principles, modularization best practices, feature flag implementation, and monorepo architecture, elevating the team's overall code quality and architectural design skills.
- Conceived and executed an automated system for cloning and building source code across a multi-repo setup, significantly reducing version conflicts and enhancing code visibility for efficient debugging.

Technologies:

- Kotlin Coroutines / Mockk, Mockito, jUnit, etc.
- Agile, Sprint, Lane based branching strategy, CI/CD
- Atlassian Jira, Atlassian Confluence, Gitlab, Jenkins, Slack, Branch.io, inVision, Dynatrace, etc.

Staff Android Developer

at iFood (via Letgo) | Oct. 2020 - Mar. 2021 | Barcelona, Spain
iFood is a leading food delivery platform, dominating markets in Brazil, Colombia, and Mexico. It's the market leader in Latin America with 50+ million Android app installations. Mission-critical achievement: Initiated and executed a strategy to refactor legacy code out of the root module, resulting in a 55% decrease in incremental build time and improved overall system architecture.

Senior Android Developer

at Letgo: Leading Classified Marketplace | Oct. 2019 - Mar. 2021 | Barcelona, Spain
One of the largest classified marketplaces in the United States at the time. Market leader in Turkey with 15+ million Android app installations. Mission-critical achievement: Optimized feed loading performance, reducing launch time by over 60 seconds.

Lead Android Developer

at Sheypoor: Top-Tier Classified Marketplace | Jan. 2018 - Jan. 2019 | Tehran, Iran
Impressive user base: 13+ million overall users, 1.5+ million daily active users. High engagement: 344+ million sessions. Key Achievement: Spearheaded initiatives that tripled daily active users from 500,000 to 1.5+ million within one year.

Senior Android Developer

at Myket: Top-Tier App Store and Social Network | Sep. 2015 - Dec. 2017 | Tehran, Iran

Java and C# developer

at Soha System | Mar. 2015 - Aug. 2015 | Tehran, Iran

Public Contributions

- 2026 tabular-center (tabula.center): State machines declared as a transition matrix, in Rust, Kotlin, Swift
  Incomplete machines fail to compile; one conformance suite runs in all three languages in CI
- 2026 torch-type-nn (github): PyTorch neural network that grows and prunes its own structure
  Matches the C reference to 1e-12; CPU and CUDA backends; reproducible tests via Nix flakes
- 2026 MFFT benchmark: A benchmark among matmul implementations with focus on MFFT
- 2025 Post: Type Mechanics
- 2025 Post: Geometrical Probability
- 2024 Post: Matrix Fast Fourier transform(MFFT) - An optimization for Matrix Multiplication
- 2024 C Numbers: This is a collection of number types, including floating numbers and p-adic numbers.
- 2024 Post: Not All codes are liabilities
- 2024 City of Toronto RAG GPT: The project received the second position award in the TMLS Hackathon.
- 2024 dragon detector: A simple ML model to detect dragon in images
- 2024 Nix Effect pod: Modularize effects of all kinds, and stay in-control, with NixOS and podman.
- 2023 Contributing to nixpkgs/androidenv: Nix is a language that is designed to have reproducible output.

Education

Sharif University Of Technology	M.Sc	in Physics	Feb. 2010	Tehran, Iran
Payame Noor University	B.Sc	in Physics	Jul. 2007	Mashhad, Iran

Tools / Skills

Developer Platform / CI/CD

GitLab CI, GitHub, Jenkins, BuildKite, Fastlane
Merge trains/tiered MR pipelines/trunk-based release
Develocity (Gradle Enterprise)/build caching
Self-hosted runners on AWS EC2
Nix flakes/Podman/Docker/custom containers
GitLab SAST/Detekt/Konsist

AI Coding Agents / LLMs

Multi-agent workflows (coder/reviewer loop)
Ticket-to-merge-request automation in CI
Daily use of AI coding tools
RAG (TMLS Hackathon), PyTorch, CUDA

Programming Language

Kotlin (advanced)/Python/Rust/Java/Swift
Scala/Groovy/Dart/JavaScript/Ruby/C

Kotlin Multiplatform/KMP

Compose Multiplatform/Android/iOS/Desktop
Advocate for independent UI/UX from platform
Main Contributor of Log4K
Main Contributor of LifecycleHandler

Android

MVVM/Android Architecture Components/JetPack
Anvil/Dagger 2/RxJava 2/Retrofit 2/...
Kotlin Multiplatform/Kotlin Coroutines
custom Kotlin Compiler Plugins/...
custom KSP plugins/...
Jetpack Compose/View Binding/...
Contributor of NixOS's Android Env

iOS

Swift/SwiftUI

Software Architecture

Clean Architecture/Hexagonal Architecture
Unidirectional data flow Architecture
Stateless Architecture
Logic Tree Architecture
MVVM/MVI/MVP/RMVVM

Principle

Clean Code/SOLID
Object Oriented Programming
Functional Programming
Isolation/Reproducibility
Stable Dependencies Principle
Stable Abstractions Principle

Design Pattern

Factory/Builder/Singleton/...
Adapter/Facade/Proxy/Visitor/Delegate/...
Strategy/Chain of responsibility/...

Modularization Strategies

Applying DIP/Decrease incremental build time
Code isolation to reduce dependencies

Android Performance Optimization

memory management/threading/profiling tools

Android Security

encryption/secure networking/permissions model

Android Test and Verification

Unit/Integration/Mockito/Espresso/JaCoCo/...
Detekt plugins/Konsist

Android Build System

ProGuard/R8/custom Gradle plugins

Build Tool

Nix/Gradle/Cargo/Sbt/Maven/Ant/Make/...

Command Line Tool

tmux/bash/zsh/git/grep/sed/netstat/awk/podman/...
adb/sdkmanager/avdmanager/xcodebuild/...

Rust

actix/rusqlite/tokio/thiserror/crossbeam
tonic/prost/openssl/flatbuffers/thrift
Rc/Arc/Mutex
FFI

Podman/Docker

Buildkit/Compose/Cache

Nix

NixOS/Nix Darwin/Home Manager
mkDerivation/mkShell

Workflow

Trunk Based development/Github flow/Git Flow
Cross-functional team

Operating System

Linux(NixOS/Fedora/Ubuntu/Debian), MacOS

Awards and Recognition

- 2024 First position in National Health Datathon
- 2024 Second position in AWS Summit's Gameday
- 2024 Second position in Toronto Machine Learning #TMLS Hackathon
- 2018 Received result-oriented badge in Sheypoor
- 2007 Ranked 44th among more than five thousand participants in the National Graduate Entrance Exam in Physics

Language

English: Advanced
Farsi: Native

The latest version of this resume is available in github
Built on October 19, 2026